



CCSDS RECORD CCSDS 231.0-B-4 CORRIGENDUM 1

Issue Date: July 2026

CORRIGENDUM 1

The Management Council of the Consultative Committee for Space Data Systems (CCSDS) has authorized the publication of technical corrigendum 1 to CCSDS 231.0-B-4, issued July 2021.

Section 5.2.4.2

- Add sentence “with probability $\geq 1 - 10^{-4}$ ” at end of first paragraph.
- Replace first line of binary sequence with

**01100011 10100001 11101101 01110010 11000110 10101100 01111001
11100010**

- Replace first part of HEX pattern with **63A1 ED72 C6AC 79E2**
- Replace “fail to decode” with “decode with probability $\lesssim 10^{-4}$ ”
- Add NOTE 3

“If the tail sequence is detected with the LDPC decoder, it will attempt, and fail, to decode the derandomized version of the tail sequence. In hexadecimal, that is 9C987328 AE457F17 39DC7AF4 640B5D95.”

Resulting in the update as shown below:

TECHNICAL CORRIGENDUM 1 TO CCSDS 231.0-B-4

5.2.4.2 Tail Sequence for LDPC (128, 64) Coding

The CLTU Tail Sequence field optionally applicable for LDPC (128, 64) coding shall be a 128-bit pattern that may be used to delimit the end of a CLTU. The sequence is constructed specifically to be a non-correctable pattern which stops the decoding process.

The Tail Sequence shall consist of the following 128-bit pattern: ^{with probability $\geq 1-10^{-4}$.}

01100011 10100001 11101101 01110010 11000110 10101100 01111001 11100010
~~01010101 01010101 01010101 01010110 10101010 10101010 10101010 10101010~~
01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101

↑ BIT 0 ↑ BIT 127

NOTES

63A1 ED72 C6AC 79E2

1. In hexadecimal, this pattern is ~~5555 5556 AAAAA AAAAA 5555 5555 5555 5555~~
decode with probability $\leq 10^{-4}$
2. A conventional LDPC decoder will reliably fail to decode when it over-runs the end of a CLTU, and this property may be used to stop the decoding process without the need for a Tail Sequence. Different decoding algorithms may not have this property, and the corresponding decoders may be used if the Tail Sequence is transmitted at the end of each CLTU. Subsection 8.2 specifies for managed parameters relevant to the Tail Sequence.
3. If the tail sequence is detected with the LDPC decoder, it will attempt, and fail, to decode the derandomized version of the tail sequence. In hexadecimal, that is 9c987328 ac457f17 39dc7af4 640b5d95.